Murilo Corato Zanarella murilocorato@gmail.com • murilocorato.github.io

employment

1. Johns Hopkins University, J.J. Sylvester Assistant Professor 2024-present

education

2. Massachusetts Institute of Technology, PhD in Mathematics 2019-2024
   Advisor: Wei Zhang
1. Princeton University, B.A. Mathematics 2015-2019

research interests

I am interested in arithmetic aspects of the (relative) Langlands program, particularly in the role of Shimura varieties, and applications to the Beilinson–Bloch–Kato conjectures, Euler systems and Iwasawa theory.

papers

8. Relative Satake transforms and arithmetic level raising for unitary Friedberg–Jacquet periods 2026
   in preparation
7. Potential maps beyond the strictly semistable case 2026
   in preparation
6. Unequal-parameter Kostka–Foulkes polynomials of type C_n at fundamental weights 2026
   preprint, `https://arxiv.org/abs/2609.39095`
5. First explicit reciprocity law for unitary Friedberg-Jacquet periods 2026
   submitted, `https://arxiv.org/abs/2602.09831`
4. Spherical functions on symmetric spaces of Friedberg-Jacquet type 2023
   submitted, `https://arxiv.org/abs/2311.00148`
3. Spherical functions of symmetric forms and a conjecture of Hironaka 2023
   submitted, `https://arxiv.org/abs/2311.00147`
2. On Howard's main conjecture and the Heegner point Kolyvagin system 2019
   Undergraduate senior thesis, preprint, `https://arxiv.org/abs/1908.09197`
1. A proof of Kolyvagin's conjecture via the BDP main conjecture 2019
   Undergraduate junior paper, preprint, `https://arxiv.org/abs/1909.07835`

invited talks

11. Massachusetts Institute of Technology (MIT): Number Theory Seminar November 2026
    Bipartite Euler systems and local harmonic analysis
10. 8th Nisyros Conference on Automorphic Representations & Related Topics July 2026
    Bipartite Euler systems and local harmonic analysis
9. Algebra and Number Theory Day March 2026
   Bipartite Euler systems and local harmonic analysis
8. Columbia University: Number Theory Seminar February 2025
   1st reciprocity law for unitary FJ periods
7. Duke University: Number Theory Seminar October 2024
   1st reciprocity law for unitary FJ periods
6. Institute for Advanced Study (IAS): Number Theory Seminar October 2024
   1st reciprocity law for unitary FJ periods
5. The University of Texas at Austin: Number Theory Seminar February 2024
   1st reciprocity law for unitary FJ periods
4. Junior Number Theory Days at Johns Hopkins February 2024
   1st reciprocity law for unitary FJ periods

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invited talks cont.

3. Tsinghua Sanya International Mathematics Forum (TSIMF) — January 2024
   1st reciprocity law for unitary FJ periods
2. Simon Laufer Mathematical Sciences Institute (MSRI/SLMath) — March 2023
   Arithmetic level raising and reciprocity laws
1. Joint Mathematics Meeting (JMM) — January 2020
   On Howard's Main Conjecture and the Heegner point Kolyvagin system

organizing

3. Number theory seminar (JHU, co-organizer) — Fall 2025 - Spring 2027
2. Number theory learning seminar (MIT, co-organizer) — Fall 2024
1. Learning seminar on Euler systems (MIT) — Fall 2022

professional activities

6. Refereed for Acta Mathematica
5. JHU reading course mentor — Spring 2026
   Mentored an undergraduate on a first course in modular forms
4. JHU reading course mentor — Spring 2025
   Mentored an undergraduate on the arithmetic of elliptic curves
3. High School Enrichment Program Teacher — 2021-2024
   Administered virtual classes with students from my former high school on undergraduate-level topics in number theory
2. MIT Directed Reading Program mentor — Winter 2022
   Mentored 2 undergraduates on analytic number theory
1. MIT Directed Reading Program mentor — Winter 2020
   Mentored 3 undergraduates on modular forms and elliptic curves

teaching

At Johns Hopkins

14. Number Theory I (110.617) (graduate level) — Fall 2026
13. Calculus II (for Physical Sciences and Engineering) (110.109) — Spring 2026
12. Representation Theory (110.422) — Fall 2025
11. Number Theory I (110.617) (graduate level) — Fall 2025
10. Calculus III (110.202) — Spring 2025
9. Number Theory I (110.617) (graduate level) — Fall 2024

At MIT (as Teaching Assistant)

8. Algebra I (18.701) — Fall 2023
7. Differential Geometry (18.950) — Fall 2023
6. Multivariable Calculus (18.02) — Fall 2022
5. Matrix Methods in Data Analysis & Machine Learning (18.065) — Spring 2021
4. Algebra I (18.701) — Fall 2021
3. Linear Algebra (18.700) — Fall 2021
2. Algebra II (18.702) — Spring 2020
1. Real Analysis (18.100A) — Fall 2020

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awards

8. Professor Joel Dean Award for Excellence in Teaching — 2025
   Awarded for extraordinary performance in teaching undergraduates.
7. Frank and Brennie Morgan Prize (hon. mention), AMS/MAA/SIAM — 2020
   Awarded for outstanding research in mathematics by an undergraduate
6. The Middleton Miller '29 prize — 2018
   Awarded for the best independent work in mathematics
5. Peter A. Greenberg '77 Memorial Prize — 2018
   Awarded for outstanding accomplishments in mathematics
4. Putnam examination — 2016-2018
   N1 prize (6th-14th) in 2016 and 2018, Honorable mention in 2017
3. Shapiro prize for academic excellence — 2017, 2018
   Award for outstanding academic achievement
2. The Class of 1861 prize — 2017
   Awarded to the sophomore with the best record on the Putnam
1. International mathematics olympiad — 2014-2015
   Silver medals in 2014 and 2015